

### Project Brief

#### Introduction

Following review of the initial designs of the 3D printer filament unwind/wind machine in Semester 1 of this academic year, management within the Engineering department of ATU Sligo have decided to continue development of the project and prepare the machines' design for manufacturing and retail.

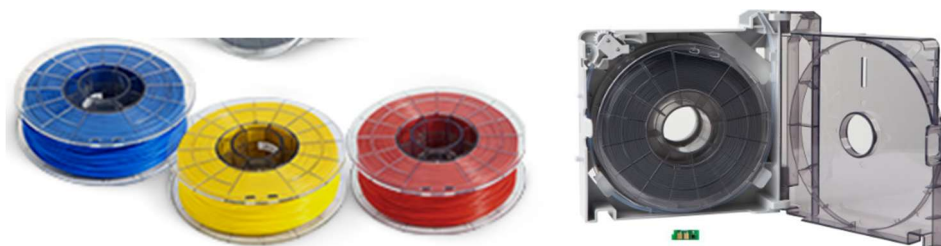
During the review of the current proposed design there are several important design areas that must be addressed to ensure the projects' success when the design phase is completed.

- Compliance with relevant regulatory standards and directives.
- Design for Manufacture and Assembly.
- Design for Reliability and Maintenance.
- Design for Sustainability.
- Prototyping and Virtual testing/simulation to ensure mechanisms work effectively.

Additionally, further enhancements must be made to improve the end user experience when operating the machine.

- Design of an inclusive, lightweight and portable workbench/trolley for the machine to sit on top which can securely hold and segregate 4 full and 4 empty filament spools, and house a 600x600x300mm standard electrical enclosure. The inclusive design must include a suitable mechanism for the working height of the machine to be adjusted vertically to suit the users' individual requirements.
- Integrate a suitable low voltage electrical device which can automatically determine when the machine shall stop running when either spool is full/empty. A standard audible alarm and /or light signal must be located on the machine to be used to alert operator to the stoppage.

Finally, Engineering Management has requested that appropriate media must be created which will present and showcase the machine designs to potential investors at the ATU Sligo Engineering Expo and thereafter.



**Figure 1: ATU Sligo Filament Spool & Cartridge (empty & full)**

### Semester 2 Filament Spool Re-wind Machine Project Guidelines

#### Requirements in this Project:

##### 1. Design:

- Review and re-design filament spool re-wind machine from Semester 1. This re-design should **incorporate all design related points outlined above**.
- **Design and build a working prototype** of a mechanism that you have included in your design. Scope of prototype to be agreed upon with lecturers.

##### 2. Planning & Organisation:

You are required to complete an array of organisational tasks, which will contribute to the project success and contain elements that may be included in your recorded report and Expo poster, including the following:

- Planning details and scheduling required before undertaking design.
- Research evidence of the specific topics applied to the project design.
- Project management plan, indicating the timeline of events in the project for all tasks / sub-tasks.
- Reach the main milestones that will apply to each group and sub-team
- Informal weekly project updates given to lecturers either in class or on digital collaborative workspace. (i.e. what has been completed in the past week, what is planned for the coming week ahead, what task is assigned to who)

##### 3. Design Media

- As some investors may be located overseas you will be required to create a **recorded report** exhibiting your group research, project management, prototyping methods and overall design using PowerPoint as the presentation tool. Guidelines and support on a suitable approach are available on the moodle page.
- **Final Design Project poster** to be created to exhibit your Semester 2 project activities and overall design at the ATU Sligo Engineering Expo.
- **Physical mechanism prototype and/or Extended Reality virtual model** to exhibit your project design alongside your poster at the Sligo Engineering Expo.

#### Design constraints / guidelines

- ATU Sligo 3D printer spools and cartridge sizes
- Specified 3<sup>rd</sup> party filament spool
- One single electric motor to provide spool rotation and filament guide motion.
- Cantilevered spool mounting shafts and tool-less fasteners to provide ease of changeover
- Reciprocating filament guide
- Filament tension control mechanism
- Standard components and fasteners to be used where applicable.
- Design for Excellence methodology to be used during project design process.

### Electric Motor

Oriental Motor 4IK25GN-CW2TE / 4GN120RA to be coupled/connected to the spool supporting shafts (and the filament guide reciprocating mechanism if using only one single motor/actuator).

Single phase 25W induction motor with a 120:1 reduction gearhead attached.



**Figure 4: AC Induction Motor & Gearhead**

### Bearings & gears

There are numerous suppliers of bearings and some companies will source their bearings from central suppliers, (such as Central Bearings Supplies, Misumi, McMaster-Carr), in which you can then specify/order the type of bearing you require from different manufacturers including FAG, NSK, NTN, IKO, Nippon Bearing, SKF etc. The majority of these manufacturers allow designers to download a CAD model of the bearing in numerous file formats, including SOLIDWORKS and neutral 3D file formats (STEP, IGES, etc.). You can select your own manufacturer type and type of bearing and download as required. An example as such is in the use of SKF bearings, in which the web site can be accessed [here](#).

## **Design Engineering Project 202**

### **Submission Requirements:**

You will be required to submit/upload the completed design project, consisting of the following:

- Detailed Project Schedule. Due Week 3
- Initial prototype design complete by Week 4 (approx.). Prototype Assembled by week 6
- Final Project Poster. Due end of week 10 (TBC).
- Images/recordings of mechanism prototype fully assembled. Due week 12.
- 3D CAD model of the final machine. (eDrawing assembly, Solidworks files, etc.). Due Week 12.
- Recorded report presentation of overall project and design elements of the machine. Due Week 13.

All necessary files should all be contained in one folder. All files should be clearly named and organised into appropriate sub-folders (e.g. a separate folder for CAD, Presentation etc).

**Late submissions can be accepted; however, 5% will be deducted per day until the project is submitted.**

**Late submissions for content required for Engineering Expo will result in groups being unable to exhibit your work in person at the expo.**